Sensor faults and actuator commands in low-cost hydroponic controllers: a multi-site retrospective audit

MD. Sameer Iqbal Chowdhury[a], Mikdam-Al-Maad Ronoue[a], Md Asaduzzaman[b], Lafifa Jamal[a,*]

[a]*Department of Robotics and Mechatronics Engineering, University of Dhaka, Dhaka-1000, Bangladesh*
[b]*Department of Agriculture and Food Technology, Kyoto University of Advanced Sciences, Japan*

Abstract

Low-cost Internet of Things (IoT) controllers regulate the nutrient solution and climate of many closed hydroponic systems, so a faulty sensor reading can directly trigger a dosing or injection command. We audited how often this happens. We first replayed the reported setpoints and executable actuator directions on a 30-day log of 121,244 samples from our closed nutrient film technique (NFT) lettuce system. Under naive control, flagged readings supplied 19.1% of pH dosing commands and 10.2% of CO_2 injection commands. A range check with 32 s persistence reduced these to zero and 2.0%. We then tested the finding on 3 public datasets under a protocol frozen before their values were opened. At an independent low-cost hydroponic chamber that logged its dose requests, up to 4.5% of the nutrient requests in a trial were issued while the electrical conductivity probe read exactly zero. Replaying its firmware rule reproduced the logged nutrient requests in 100% of rows. However, range check and persistence met the pre-registered criterion in only 1 of 8 trial–channel cells, and persistent zero EC readings passed both tests. Rejecting readings at a sensor bound removed 69–99% of these commands. A commercial greenhouse showed no out-of-range or bound CO_2 readings. LiveGate, a learned consistency and liveness gate, did not outperform simple baselines under a frozen protocol. Delivered doses, prevented damage and crop effects remain unmeasured, and a dose-response model could not be validated from the logs.

*Corresponding author.

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1. Introduction

Closed hydroponic systems recirculate a nutrient solution whose pH, electrical conductivity (EC) and temperature must be held within narrow ranges [1, 2]. Automated controllers built from low-cost sensors, microcontrollers and single-board computers now regulate these quantities using dosing pumps, valves, chillers and coolers [3, 4]. In such controllers, a sensor reading is not only a data point, since it can trigger a physical action. A single spurious high pH reading can authorize acid dosing into the whole recirculating volume. Therefore, knowing how often a controller acts on flagged readings is as important as knowing how often the readings are flagged.

Sensor-fault studies characterize faults and their detection [5, 6, 7, 8], while studies of IoT hydroponic systems describe architectures and crop outcomes. However, the number of executable commands supplied by flagged readings is rarely reported. Answering this question requires accounting for actuator direction, persistence, cooldown and the age of held readings together. It also requires checking whether a finding from one system holds in others.

In this paper, we audit this question on our deployed closed NFT lettuce system and test it on 3 public datasets. The contributions of this paper are as follows:

1. We provide an executable-command replay that attributes each authorization to its source reading, and apply it to a 30-day hydroponic log under 6 gating rules.
2. We report a pre-registered multi-site replication, including an independent chamber whose logged dose requests validate the replay and show nutrient dosing on zero EC readings.
3. We report a frozen comparison of LiveGate with 14 alternatives, with its negative overall result, and a failed attempt to identify a dose-response model from logged requests.
4. We derive design checks for gating and instrumentation, with explicit limits on physical and agronomic inference.

2. Related Work

Automated hydroponic control typically measures pH and EC in the nutrient tank and corrects them with acid, base, concentrate or water dosing, often with a mixing delay between doses [3]. IoT variants add networked logging and remote monitoring [4]. Field sensor deployments exhibit outliers, spikes, stuck-at values, offsets, drift and missing data [5, 6], and reviews of agricultural IoT report similar fault classes [8]. Classical detectors include robust outlier filters [9], the Cumulative Sum (CUSUM) [10, 11], Exponentially Weighted Moving Averages (EWMA) [12], the Kalman filter [13] and the Local Outlier Factor (LOF) [14]. A survey of model-based fault diagnosis is given in [15]. However, range and local-change tests can accept a plausible constant trace, and persistence-based confirmation can then reinforce an erroneous authorization.

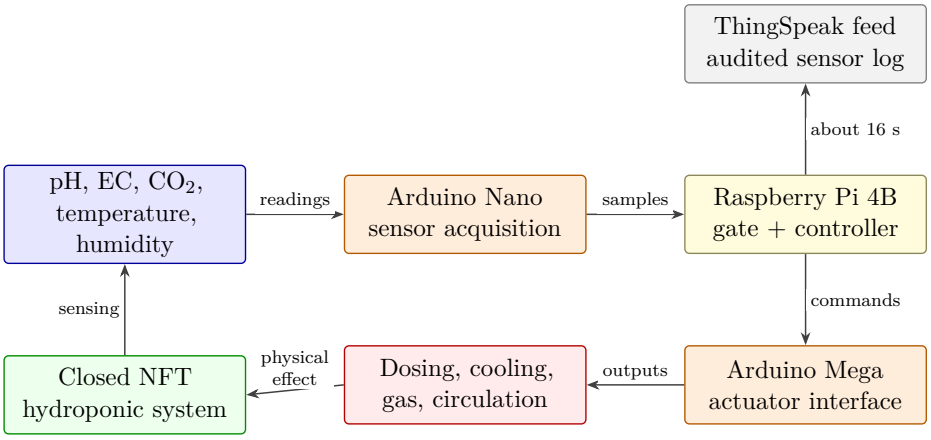
Ferentinos et al. [16] designed and developed neural-network detectors for mechanical, sensor and biological faults in deep-trough hydroponics, using actuator control signals among their inputs. Ferentinos and Albright [17] also proposed a real-time supervisor for the same setting. Lin et al. [18] designed SensorTalk, which feeds the readings of a standard sensor to the actuators during calibration. More recently, Karimzadeh et al. [19] addressed EC and pH fault diagnosis in hydroponic systems, and Pompeo et al. [20] applied outlier detection to assess the stability of indoor farming systems. Munser et al. [21] isolated faults with parity residuals and observers in controlled-environment agriculture. Bontsema et al. [22] used reference-measured greenhouse sensor errors to simulate their effects on production and energy use.

These studies detect, isolate or simulate faults, which requires different information and endpoints from retrospective command attribution. Our contribution is an authorization audit with explicit source provenance and executable directions, tested across sites. It is not a new general principle of fault diagnosis, nor evidence of measured physical consequences.

3. System and Deployment

Our system is a closed NFT system for leaf lettuce, which consists of a Growth Module (GM) and a Nutrient Solution Container Module (NSCM). Figure 1 shows its sensing-to-actuation path along with the deployed system, and Table 1 lists the controlled quantities. The GM measures air temperature and humidity with a DHT22 sensor, and CO_2 with an MH-Z19C

non-dispersive infrared sensor whose range ends at 5000 ppm. The NSCM measures pH and EC with Atlas Scientific analog kits, along with water temperature and water level. The sensors are read by an Arduino Nano with median filtering, and the actuators are driven by an Arduino Mega. A Raspberry Pi 4 Model B runs the controller and uploads the readings to ThingSpeak at intervals of approximately 16 s. However, actuator commands and relay states were not logged. As such, every command from our system is reconstructed from the sensor log.



Command and relay states were not logged.



Figure 1: Sensing-to-actuation path (top) and our deployed NFT system (bottom). Sensor readings were logged, while command and relay states were not recorded.

The description of our system specifies a correction whenever a reading leaves its setpoint band, followed by a 10-min mixing lockout after pH or EC dosing. The replay uses this interval as a common cooldown for all 5 channels, and its extension to CO_2 and cooling is an assumption. In the first season, the controller acted on readings without further gating. For

Table 1: Controlled quantities, deployed setpoint bands and executable actuator directions. Humidity and water level were monitored but not actuated.

Quantity	Setpoint band	Actuator	Executable direction
pH	5.8–6.5	Acid / base dosing	Both
EC	1000–1500 $\mu S\,cm^{-1}$	Water / concentrate dosing	Both
CO_2	1900–2100 ppm	Solenoid valve on cylinder	Raise only
Water temperature	18–22 °C	Refrigerative chiller	Lower only
Air temperature	20–26 °C	Thermoelectric cooler	Lower only

the second season, we added a sensing rule that accepts a reading when it differs from the last trusted value by at most $(1 + \beta)\xi$, with a reset after 50 consecutive rejections. The description of its actuation bands does not uniquely specify the firmware behavior. This led us to build a legacy-gate proxy that retains the sensing rule and reset, omits the actuation bands and requires 16ψ seconds of persistence, with ξ and ψ re-estimated from development data. Thus, this arm is an approximate comparison and not a reconstruction of the firmware.

The analysed deployment ran from 27 February to 28 March 2024 in Dhaka, Bangladesh, at ambient temperatures of approximately 25–36 °C. We grew 7 lettuce seedlings in each of 3 setups: our automated NFT system, pots with soil and compost, and a manually managed Deep Water Culture (DWC) system.

4. Methods

4.1. Fault Flags

The log contains 121,244 rows with a median interval of 16 s. Water level is excluded, since its raw values were never calibrated. For each channel, we flagged 3 disjoint categories of readings:

1. **Out-of-range:** Readings outside the physical range of the sensor, such as pH outside 0–14 or CO_2 outside 250–5000 ppm.

2. **Bound:** Readings exactly at a declared physical bound.
3. **Spike:** In-range readings more than 8 robust standard deviations from a centred 9-sample median.

We also counted flat-lines of identical readings lasting at least 1 h, and logging gaps longer than 120 s. These are descriptive flags and not ground-truth fault labels, since a bound reading may be a true excursion and a spike may be a legitimate transient.

4.2. Fault-to-Actuation Audit

The audit replays the setpoint bands and the 10-min cooldown on the unmodified log under 6 gating arms: (1) a naive threshold, as in the first season, (2) a physical-range check, (3) a range check with 32 s persistence, (4) the legacy-gate proxy, (5) arm 3 with the LiveGate liveness test and (6) arm 3 with rejection of bound readings. The last accepted value may be held for at most 64 s, and a gap longer than 120 s invalidates it. Directions that the installed actuator cannot execute are removed before persistence and cooldown. Each command is attributed to the reading that supplied its held estimate. The replay uses a fixed historical background, so it measures differences in authorization and not actual dosing or crop effects. After reproducing the audit, we declared and ran a sensitivity grid over persistence (16–96 s), held age (32–128 s), cooldown (300–1200 s) and spike cutoffs (6–10 scales), giving 1620 cells.

4.3. Controlled Benchmark and LiveGate

Faults that cannot be identified from the log alone were studied by injection into 240-sample windows drawn from the validation and evaluation blocks (the second and third 30% of the log). Each window received 1 of 12 scenarios on 1 channel, covering spikes, biases, drifts, a stuck value, saturation at full scale, a dropout and legitimate level shifts. A common controller then holds, confirms and cools down the accepted readings. Coverage is the fraction of reference opportunities from the clean background that are matched within 180 s, and undesirable authorizations are commands issued when the background is in band or in the opposite direction. Each method was tuned on validation windows at coverage targets of 0.5, 0.8 and 0.95, and then scored once on the evaluation windows. The evaluation windows span 7 source days, so paired 95% intervals resample those days with 1000 bootstrap replicates [23].

We developed LiveGate to accept a reading y_t only if it is consistent with a learned forecast and the sensor appears alive. The forecast adds a Huber-regression term [24] to the mean ℓ_t of the last 5 accepted readings, using the trend, the hour of day and the other 4 actuated channels. The score $s_t = |y_t - \hat{y}_t|/\sigma$ is compared with a split-conformal threshold [25, 26], given by Equation 1:

$$q_\varepsilon = s_{(\lceil (n+1)(1-\varepsilon) \rceil)} \tag{1}$$

$s_{(k)}$ is the k-th smallest of the n clean calibration scores. We make no formal coverage guarantee, since the scores are dependent. The liveness test rejects a reading when the smoothed survival $S(r_t)$ of the current run length r_t of identical readings falls below ε, as given by Equation 2:

$$S(r) = \frac{|\{\text{runs} \geq r\}| + 1}{N + 1} \tag{2}$$

N is the number of training runs, excluding readings at a declared bound. The single parameter $\varepsilon \in \{0.002, 0.01, 0.05\}$ sets both thresholds. We committed the method and the decision rules before running LiveGate on the measured log. As we ran the synthetic unit tests, we noticed cascading rejections, which led us to anchor the forecast on the 5-sample level instead of the last value. The data had been analysed in earlier studies, so the evaluation is retrospective.

4.4. Multi-site Replication

To test whether the audit findings hold beyond our system, we wrote and committed a replication protocol before parsing any numerical values from the public datasets. Three sources were retained:

1. **S1:** an indoor hydroponic chamber described by Pompeo et al. [20], with 4 trials (T1–T4) of 4 nutrient lines each. Atlas Scientific probes measure pH and EC, and the firmware logs each acid and nutrient dose request at intervals of approximately 10 s. Its published firmware requests acid when pH exceeds 5.85 and nutrient when EC falls below 1100 µS cm^{-1}, with no persistence or cooldown.
2. **S2:** the second Autonomous Greenhouse Challenge [27], with 6 commercial glasshouse compartments logging CO_2 concentration, computed CO_2 dosing and setpoints at 5-min intervals.
3. **S3:** an aquaponic IoT logger with low-cost pH, total dissolved solids (TDS) and water-temperature sensors [28]. It has no controller, so it was used for fault prevalence only.

One further candidate was excluded because its published sensor data had already been imputed and cleaned of outliers. The flag definitions were inherited unchanged, and timing was scaled to each sampling rate, with persistence of 2 samples and a maximum held age of 4 samples. The protocol defined 4 endpoints:

1. Fault prevalence at every site.
2. The fraction of logged dose requests (S1) or dosing intervals (S2) whose same-row reading was flagged.
3. The agreement between the replayed firmware rule and the logged requests (S1).
4. The flagged commands under naive, range-check, range-plus-persistence and bound-rejection replay (S1).

The finding was declared replicated in a trial–channel cell when range plus persistence cut flagged-source commands by at least 90% relative to naive replay while keeping at least 80% of the unflagged commands. Cells with fewer than 10 naive flagged commands were not testable. The 4 lines of a trial share 1 chamber and logger, so they were pooled within each trial, giving 8 cells. The bound-rejection arm was declared and reported, but the criterion was not set for it.

Finally, we tested whether a dose-response model could be identified from the S1 logs, which would allow the physical effect of flagged requests to be estimated. Under a second frozen protocol, we fitted a first-order model of EC change per 1-min bin, driven by logged nutrient requests and water top-ups, on trials T1–T2. We then simulated EC open-loop on T3–T4. The model was accepted only if it beat holding the last value by 20% at a 30-min horizon in at least 6 of 8 test cells.

5. Results

5.1. Faults and Regulation in Our System

Table 2 lists the fault inventory of our 30-day log, and Figure 2 shows the daily flags. Our observations are as follows:

1. pH produced 464 physically impossible readings, with values up to 67,101. This is consistent with corrupted analog or serial transfers, although the fault location cannot be diagnosed from the log.

2. CO_2 produced 819 out-of-range readings and 961 spike flags. Furthermore, 2.3% of its samples were exactly at the 5000 ppm full scale, including one episode lasting 8.5 h.
3. Humidity read 99.9% for 99.97% of samples. Saturation and a stuck channel cannot be distinguished without independent measurements.
4. All channels shared 194 logging gaps longer than 120 s, which together covered 23% of the deployment.

Table 2: Fault inventory of our 30-day deployment log, with 121,244 readings per channel

Channel	Out of range	Maximum out-of-range	Spikes	Flat-lines ≥1 h	Longest (h)
pH	464	67,101	76	0	–
EC	0	–	14	0	–
CO_2	819	200,850	961	3	8.5
Air temperature	1	99.9	17	1	1.4
Water temperature	0	–	39	7	1.5
Humidity	0	–	10	79	35.2

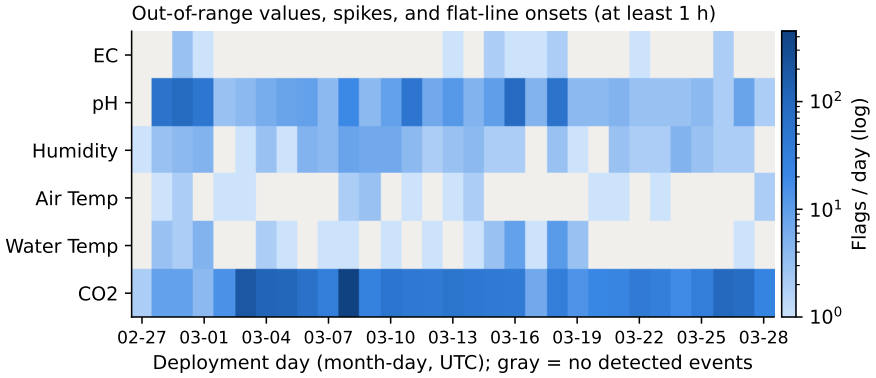


Figure 2: Daily descriptive flags in our log: out-of-range readings, isolated spikes and onsets of flat-lines lasting at least 1 h

Figure 3 shows the longest CO_2 full-scale episode, which was preceded by a steady rise from approximately 2000 to 5000 ppm over 2 h. The true concentration may have exceeded the sensor range, such as through a valve that failed open, but reference and valve-state measurements are absent. Among

valid readings, 88% of pH, 84% of EC and 47% of CO_2 readings were in band. Only 3% of valid air-temperature readings (median 28.6 °C) and 10% of valid water-temperature readings (median 24.8 °C) were in band. This indicates poor temperature regulation, consistent with insufficient cooling capacity in the hot season, although actuator failure and calibration error cannot be ruled out from the log.

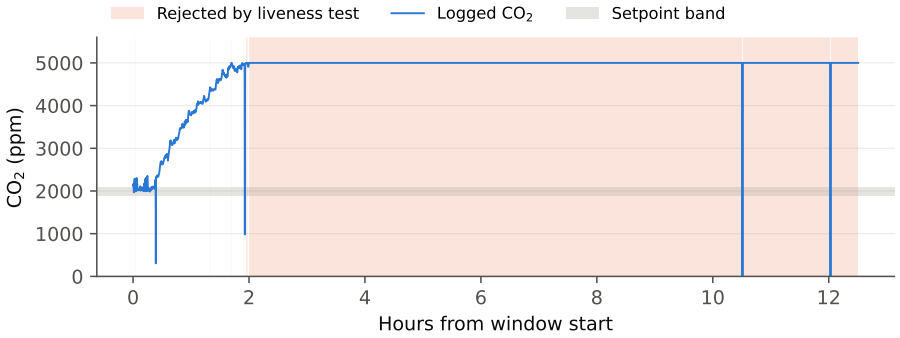


Figure 3: Longest CO_2 full-scale episode in our log, with readings rejected by the replayed liveness test ($\varepsilon = 0.05$) shaded

5.2. Fault-to-Actuation Audit of Our System

Table 3 lists the executable replay commands under each gating arm, and Figure 4 shows the share that used flagged readings. Our observations are as follows:

1. Naive replay issued 655 pH dosing commands, of which 125 (19.1%) used flagged readings, and 2149 CO_2 injection commands, of which 219 (10.2%) used flagged readings.
2. A range check left 40 flagged pH commands, made up of 28 bound readings and 12 spikes.
3. Adding 32 s of persistence removed all flagged pH and EC commands, and reduced flagged CO_2 injections to 9 of 456 (2.0%), all heuristic spikes.
4. The 5000 ppm full-scale CO_2 readings requested no injection, since that loop can only raise the concentration. As such, liveness and bound rejection gave the same CO_2 counts as range plus persistence.

5. The legacy-gate proxy issued 43–72% as many commands as range plus persistence. Nevertheless, it authorized 13 CO_2 injections from out-of-range readings, since its reset accepts the current value after 50 rejections.

Our initial audit discarded unavailable directions only after they had consumed cooldown. Upon investigation, we learned that this let an unavailable direction delay later executable commands. This led us to remove such directions before authorization, which changed the naive CO_2 counts from 1010/68 total/flagged to 2149/219, while the pH and EC results were unchanged.

Table 3: Executable replay commands in our system (Tot.) and commands using flagged source readings (Flag.). No actual command execution was logged.

Gating arm	pH dosing		EC dosing		CO_2 injection	
	Tot.	Flag.	Tot.	Flag.	Tot.	Flag.
Naive threshold	655	125	535	6	2149	219
Range check	593	40	535	6	2105	148
Range + persistence	417	**0**	526	**0**	456	9
Legacy-gate proxy	216	**0**	227	**0**	330	13
Range + persistence + liveness	417	**0**	526	**0**	456	9
Range + persistence + bound	417	**0**	526	**0**	456	9

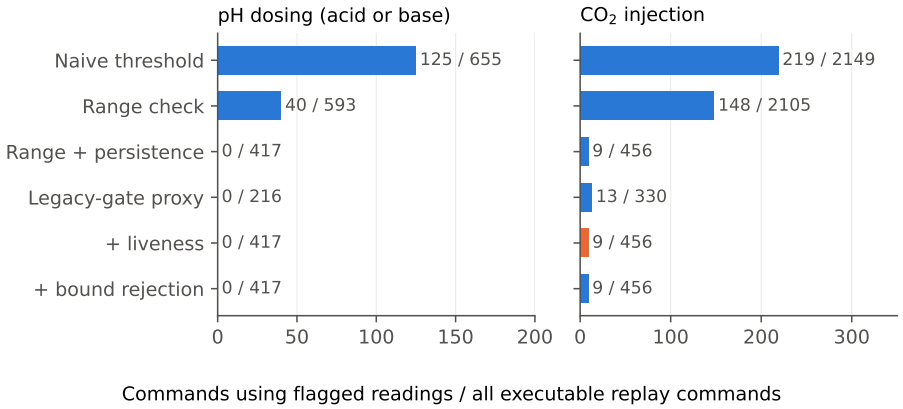


Figure 4: Executable replay commands using flagged readings in our system, relative to all commands under each arm

The sensitivity grid falsified our stronger prediction that range plus persistence would remove every flagged pH command across timings. There were 0 to 3 flagged pH commands per setting, with nonzero counts in 54 of 108 cells, while EC remained at zero throughout. Relative to naive replay at the same timing, flagged counts fell by 98.1–100% for pH and by 87.1–99.7% for CO_2. Thus, the zero-count pH result is specific to the original setting.

5.3. Multi-site Replication

Table 4 lists the EC results at S1, where the firmware rule could be verified. Our observations are as follows:

1. Replaying the published firmware rule reproduced the logged nutrient requests in 100% of rows, with a single mismatch in 277,844 rows. For pH, agreement was only 56.5–99.4% per line, so the pH rule could not be verified and its replay is reported as unverified.
2. The logged requests show that flagged readings drove real dose requests. In T3, 5003 of 110,276 nutrient requests (4.54%, day-cluster 95% interval 0.06–17.8%) were issued while a probe read exactly 0 $\mu S\,cm^{-1}$. For acid dosing, 0.04–0.55% of logged requests used flagged readings.
3. Range plus persistence met the replication criterion in only 1 of 8 cells (T2 pH), with 6 cells failing and 1 not testable. For EC, it removed only 1–69% of flagged-source commands, since a zero reading lies inside the physical range and persists.
4. Rejecting readings at a sensor bound removed 69–99% of flagged EC commands, while keeping 87–98% of unflagged commands in T1–T3. In T4, EC hovered around the 1100 $\mu S\,cm^{-1}$ threshold, so 2-sample persistence removed 97% of the single-sample commands.
5. At the commercial site S2, no CO_2 reading was out of range or at a bound in any of the 6 compartments. Its 0.26–6.96% flagged dosing intervals were all heuristic spikes, which are plausibly real CO_2 dynamics at 5-min sampling.
6. At S1, zero EC readings occupied 0–0.45% of samples per line, and logging gaps covered 4.5–21% of elapsed time. At S3, 1.09% of TDS and 0.04% of pH readings were out of range. The gap fraction of S3 could not be estimated, since its timestamps have a resolution of 1 min.

The dose-response model beat holding the last value in 0 of 8 test cells. The fitted effect per nutrient request was -0.02 to 0.55 $\mu S\,cm^{-1}$, against 30-min EC variation of approximately 4–11 $\mu S\,cm^{-1}$. As such, the physical effect of the flagged requests cannot be estimated from these logs.

Table 4: EC nutrient dosing at the independent chamber S1, per trial with 4 lines pooled. Logged: firmware dose requests. Replay columns give flagged-source commands. Retention: unflagged commands kept by range plus persistence relative to naive replay.

	Logged requests		Replay, flagged commands			
Trial	Total	Flagged (%)	Naive	Range + persistence	+ Bound rejection	Retention (%)
T1	9355	0.24	22	12	5	86.6
T2	29,999	1.15	346	327	15	92.9
T3	110,276	4.54	5003	4953	70	97.8
T4	2260	1.15	26	8	8	2.6

5.4. Controlled Benchmark and LiveGate

Table 5 lists the locked evaluation at the 0.8 target. No method reached the 0.95 target on validation, and LiveGate did not meet the frozen criterion for the best method at 0.5 or 0.8. At 0.5, LOF had the fewest undesirable authorizations (0.075 per trial against 0.135 for LiveGate), but at a coverage of only 0.357. At 0.8, persistence only reached a higher coverage with fewer undesirable authorizations than LiveGate. An exploratory variant that floored the forecast scale at the sensor resolution gave identical results, since the conformal threshold moves with the scale.

After the locked comparison, we recognised that the liveness results had not been compared against the simplest rules aimed at the same faults. This led us to add a fixed repeat cap and physical-bound rejection as post-hoc comparators. Learned liveness reduced commands from injected stuck readings by 79.3% (paired 95% interval 71.0–88.9%) relative to persistence. However, the useful-command coverage in those trials fell from 0.982 to 0.273. Physical-bound rejection issued no commands from saturated sources, while its useful-command coverage in those trials was 0.073. Thus, both gates that strongly suppressed stuck or saturated sources paid for it in useful commands.

5.5. Crop Outcome in Our System

In the same season, our NFT plants produced a mean aerial fresh mass of 264.8 g (SD 35.0, $n = 7$), compared with 68.2 g (SD 19.5) in soil and 109.5 g (SD 23.3) in manual DWC. However, these treatments differ at once in growing medium, enclosure, CO_2 enrichment, cooling, nutrient management and automation. As such, the comparison describes the deployment context and does not isolate the effect of automation or of any gating rule.

Table 5: Locked evaluation at the 0.8 validation target. A dagger marks evaluation coverage below 0.8, and a double dagger marks a post-hoc comparator.

Method	Coverage	Undesirable / trial
Persistence only	0.886	0.117
CUSUM	0.862	0.121
Hampel / moving median	0.848	0.150
EWMA	0.857	0.150
Kalman	**0.910**	0.158
GLR	0.854	0.165
LiveGate	0.790†	0.135
LiveGate, liveness only	0.663†	0.104
LiveGate, forecast only	0.837	0.150
Fixed repeat cap‡	0.849	0.117
Physical-bound rejection‡	0.798†	**0.083**

6. Discussion

6.1. Which Faults Matter for Actuation

Under naive control, both impossible values and bound readings reached the actuators. A range check removes impossible values, and persistence removes most short spikes and brief bound readings, as in our pH channel. However, a reading that stays pinned at a bound is in range and persistent, so it passes both tests. Whether it causes commands depends on the actuator direction. In our system, CO_2 readings pinned at the 5000 ppm ceiling requested no injection, since that loop can only raise the concentration. At S1, EC readings pinned at zero lay below the nutrient setpoint of a raise-only loop, so the controller requested nutrient on every cycle. Thus, the same sensor signature is harmless in one direction and drives dosing in the other.

The replication also shows the value of logging commands. At S1, the logged requests allowed the replay to be validated exactly for EC, and they showed that fault-driven dosing really was requested. However, without delivered volumes and reference measurements, neither site can show how much nutrient was added or what it did to the crop. Our attempt to recover this from the logs failed, since a single request changed EC by much less than the normal variation.

6.2. Design Guidelines

Based on our results, we suggest the following design checks:

1. Readings exactly at a sensor bound should be rejected before persistence is applied. At S1, persistence alone let most commands from pinned zero EC readings through.
2. Gating should be evaluated with the actual actuator directions and cooldown, since the same pinned reading can be harmless or harmful depending on direction.
3. Resets that eventually accept any value after repeated rejections should be avoided. The legacy-gate proxy admitted out-of-range CO_2 readings through such a reset.
4. Run-length and bound rejection should be compared against persistence at matched coverage before learned gating is adopted, since every gate tested here traded useful commands for fewer flagged ones.
5. Commands, relay states, delivered volumes and independent references should be logged alongside the readings. Logged requests are enough to validate a replay, but not to measure its consequences.

6.3. Limitations

As with any retrospective study, our audit leaves room for future improvement. Its main limitations are as follows:

1. Commands in our system were reconstructed, and only S1 logged its requests. No site logged delivered volumes, so actual dosing, avoided damage and crop benefits are unmeasured.
2. The units within each site share an apparatus or facility, and the 4 sites differ in crop, sensors and controller. They are not independent replicates of one design.
3. The pH rule at S1 could not be verified, and its firmware may differ from the published version. S1 has no declared licence, so its raw rows are not redistributed.
4. Bound rejection was a declared secondary arm and not the preregistered criterion, so its benefit is exploratory.
5. Spike flags are heuristic and bound readings are ambiguous. The LiveGate benchmark uses fixed backgrounds, 7 evaluation days and post-hoc comparators, and our data had been analysed before.
6. The crop comparison is confounded by simultaneous differences between the treatments.

7. Conclusions

In this paper, we audited how sensor faults propagate into actuator commands in low-cost hydroponic controllers. In our system, flagged readings supplied 19.1% of pH and 10.2% of CO_2 commands under naive replay, and a range check with 32 s persistence reduced these to zero and 2.0%. A preregistered replication on 3 public datasets did not confirm this as a general rule. At an independent chamber, the logged requests validated our replay exactly for EC and showed nutrient dosing on zero EC readings, which range checks and persistence did not stop. Rejecting readings at a sensor bound removed most of these commands, and the commercial greenhouse showed no such readings. LiveGate did not outperform simple rules.

Thus, the engineering contribution of this work is a validated method for counting fault-driven commands, together with evidence that readings pinned at a sensor bound can pass both range checks and persistence. Delivered doses and agronomic consequences remain unmeasured, and measuring them requires logging delivered volumes and independent references in future deployments.

Data and Code Availability

The code, frozen protocols, our measured 30-day sensor log, harvest data and supporting results are publicly available at `https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-Hydroponics-System-with-Intelligent-Parameter-Control-and-P`. The 3 public datasets are available from their cited sources, and the repository contains the scripts that reproduce every multi-site result from them.

Declaration of Competing Interest

The authors declare no competing interests.

Declaration of Generative AI and AI-assisted Technologies

AI-assisted tools were used to help write analysis code, draft text and check consistency. The authors reviewed the content and are responsible for it.

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